

Probit Offset (market-anchored)

probit family, scored against the market on 2015-2026, 5 seed(s), from unrecorded commit

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Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Probit Offset (market- anchored) probit	MEASURED AGAINST the market the same yardstick in all four reports	SEEDS 5 7, 17, 42, 123, 256	TEST YEARS 12 2015-2026
BOOSTING ROUNDS 400 3-fold CV	TRAINED ON 2010-2014 then expanding a separate fit per test year, each on 2010 through the year before it -- 12 fits in all	SOURCE COMMIT n/a clean	

These are walk-forward figures, not one model's: for each test year the model was fitted on 2010 through the previous year and then scored that year, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window expands each year, so the last year's model saw the most data and the first year's the least. These are the incumbent Offset's margins, reused exactly rather than refitted, so the window is that run's.

Config digest unrecorded, model view bagged_probit_offset, artifacts read from results/probit_models.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use model_minus_market_log_loss: race_multinomial_log_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA VS MARKET -0.004358 negative is better	RACES 8,675	YEARS BETTER 10/11	ISOLATION DELTA -0.000943 against the comparator it was designed against
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Probit Offset (market-anchored) is better than the market by 0.004358 race_multinomial_log_loss on 8,675 races. The direction holds in 10 of 11 evaluated years. Against the incumbent Offset model on the same margins -- the comparison this architecture was designed to make, which is the only one that isolates the link change -- the delta is -0.000943. Every seed agrees on the sign of that isolation delta.

Comparator and evaluation population

The market's own implied probabilities on the same races. Every published report is scored against the market, so the four headline numbers mean the same thing and can be read against one another. For the production model the market is also the baseline by construction -- the booster learns a residual on log(market_prob) -- and for the probit candidates it is the common yardstick rather than the quantity they were designed against; the comparison each was designed against is in the technical appendix.

TEST RACES 8,675	YEARS 12 2015-2026	SEEDS 5	HONEST-SCALE YEARS 11 years whose utility scale was fitted rather than fallen back
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Some years fell back rather than fitting their own scale

2015 did not produce an honest utility scale (insufficient_history). Those years are excluded from the headline delta and from the bankroll paths, and shown separately in the year-by-year table: a fallback scale is not a measurement of the candidate.

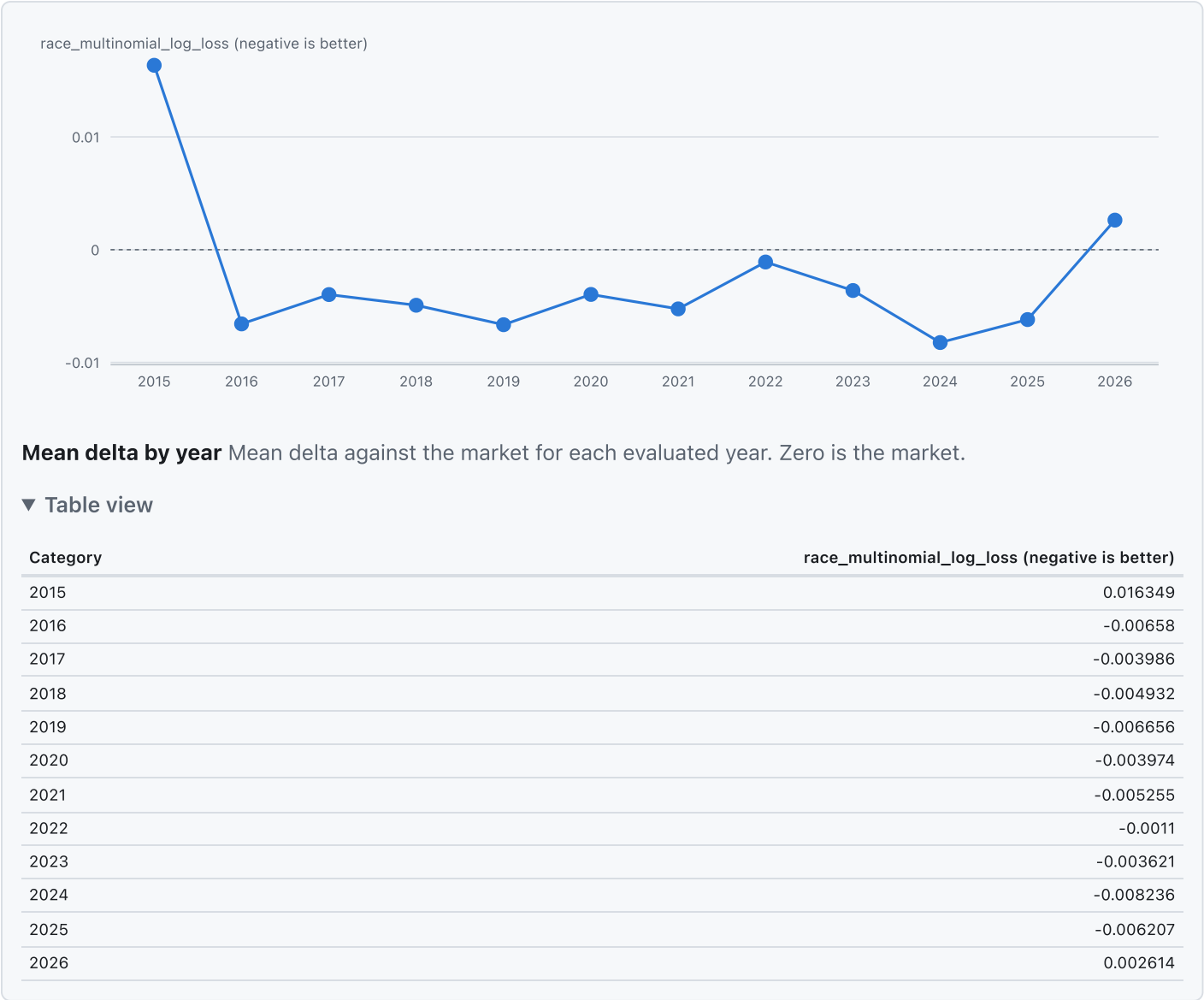
Overall probability performance

races	model log loss	market log loss	market delta	better year share
8675	2.024447	2.028805	-0.004358	0.909091

One row, pooled over every evaluated race, and measured against the market: market_delta is the model's race log loss minus the market's on the same races, so negative means the model is better. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	market log loss	market delta	honest
2015	742	2.015068	1.998720	0.016349	no
2016	744	2.092697	2.099277	-0.006580	yes
2017	772	2.017573	2.021559	-0.003986	yes
2018	756	2.056572	2.061504	-0.004932	yes
2019	759	1.926170	1.932827	-0.006656	yes
2020	799	2.006427	2.010400	-0.003974	yes
2021	783	2.001404	2.006659	-0.005255	yes
2022	783	2.050094	2.051195	-0.001100	yes
2023	796	2.020445	2.024066	-0.003621	yes
2024	792	2.017508	2.025744	-0.008236	yes
2025	748	2.051190	2.057397	-0.006207	yes
2026	201	2.028838	2.026224	0.002614	yes



Ahead of the market in 10 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: candidate and comparator are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. Fallback-scale years are excluded here, as they are from the headline.

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2016	2.092581	2.093236	-0.000655
2017	2.018646	2.018225	0.000421
2018	2.057226	2.057516	-0.000290
2019	1.928633	1.927971	0.000662
2020	2.007505	2.009214	-0.001709
2021	2.000953	2.001587	-0.000633
2022	2.050874	2.051204	-0.000330
2023	2.021237	2.021954	-0.000718
2024	2.017259	2.017686	-0.000427
2025	2.053116	2.053894	-0.000779
2026	2.029934	2.030893	-0.000959

If the calibrated delta is materially smaller than the uncalibrated one, part of the candidate's advantage was a temperature the comparator could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement compares a model's first pick against the market's. This candidate is scored against its own comparator on fixed mean utilities, not against the market, so a market top-choice comparison would answer a different question than the rest of this report.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is a choice [link](#) applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects belonging to this model to attribute. A choice link changes how utilities become probabilities; it has no features of its own. The mean utilities come from the comparator's fit, so attributing their feature effects here would credit this candidate with another model's work -- and for the market-anchored architectures those are the incumbent Offset's margins exactly, whose per-feature effects, response directions and group ablation are in the Bagged Offset report. What is specific to this candidate is the link and its utility scale, and both are in the technical appendix.

Representative races

Not applicable to this model. Representative races are chosen by how much a model's correction moved a runner's probability. This candidate applies a link to fixed mean utilities, so the movement to explain belongs to the comparator's fit and is shown in that model's report. A race chosen here would be chosen by integration noise.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
flat_stake@edge>=0.02	2655	26,550,000	2,592,000	0.097627	0.230477	0.132261	12,592,000	1746
kelly_no_rebate@0.01	3318	5,540,860	446268.500000	0.080541	0.043660	0.032880	10446268.500000	2210
kelly_no_rebate@0.02	3343	11,272,550	869024.000000	0.077092	0.083332	0.064842	10,869,024	2232
kelly_no_rebate@0.05	3350	29,435,390	2,012,729	0.068378	0.183382	0.155179	12,012,729	2239
kelly_no_rebate@0.1	3352	61,666,500	3353997.000000	0.054389	0.289231	0.288621	13,353,997	2240
kelly_settled_rebate@0.01	3319	5,587,210	448962.000000	0.080355	0.058282	0.030273	10,600,136	2211
kelly_settled_rebate@0.02	3343	11,632,560	904,457	0.077752	0.138195	0.054747	11,481,996	2228
kelly_settled_rebate@0.05	3351	32,396,920	2255565.500000	0.069623	0.351941	0.127529	14218249.500000	2234
kelly_settled_rebate@0.1	3355	75,650,170	4259989.500000	0.056312	0.657041	0.239877	19290763.500000	2238
rebate_aware_kelly@0.01	3646	8,722,790	338914.500000	0.038854	0.079050	0.038756	10822587.500000	2206
rebate_aware_kelly@0.02	3797	21,312,520	989793.500000	0.046442	0.212640	0.080152	12369387.900000	2298
rebate_aware_kelly@0.05	5038	74,758,690	1254079.500000	0.016775	0.511552	0.187908	16678771.850000	3262
rebate_aware_kelly@0.1	6573	202,278,270	1252534.500000	0.006192	0.974015	0.341975	26485578.700000	4692

Model view bagged_probit_offset. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

What this settlement can decide

Secondary diagnostic. This suite selects on paired race log loss; betting returns cannot select a model and are not evidence for or against a probit candidate.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.033997	0.033997	-0.0434	718
0.050000 to 0.100000	0.072664	0.072664	0.0151	610
0.100000 to 0.200000	0.143784	0.143784	0.0230	561
0.200000 to unbounded	0.511528	0.511528	0.3502	766

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009690	0.009690	-0.1013	-0.0979	-0.0877	-0.0766	663	688	695	697
0.020000 to 0.050000	0.033997	0.033997	0.0176	0.0170	0.0185	0.0206	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0542	0.0536	0.0528	0.0501	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1521	-0.1528	-0.1546	-0.1573	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.2149	0.2092	0.1937	0.1688	766	766	766	766

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009677	0.009677	-0.1025	-0.0939	-0.0788	-0.0586	664	688	696	700
0.020000 to 0.050000	0.033997	0.033997	0.0156	0.0163	0.0188	0.0217	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0549	0.0547	0.0556	0.0546	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1523	-0.1534	-0.1562	-0.1599	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.2150	0.2107	0.1969	0.1738	766	766	766	766

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.032218	-0.032218	-0.0699	-0.0459	-0.0344	-0.0230	327	453	1688	3227
0 to 0.020000	0.009677	0.009677	-0.1497	0.0506	-0.0037	-0.0092	664	689	695	691
0.020000 to 0.050000	0.033997	0.033997	0.0553	0.0334	0.0225	0.0021	718	718	718	718
0.050000 to 0.100000	0.072664	0.072664	0.0147	0.0384	0.0131	-0.0108	610	610	610	610
0.100000 to 0.200000	0.143784	0.143784	-0.1503	-0.1433	-0.1623	-0.1283	561	561	561	561
0.200000 to unbounded	0.511528	0.511528	0.1804	0.1787	0.1646	0.1380	766	766	766	766

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL HK\$10,000,000	PRESETS SETTLED 1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly	PATH MODES annual reset, continuous	SEEDS 5 7, 17, 42, 123, 256
TRAINED ON 2010-2014 then expanding a separate fit per test year, each on 2010 through the year before it -- 12 fits in all	BOOSTING ROUNDS 400	YEARS STAKED 2015-2026	

These are walk-forward figures, not one model's: for each test year the model was fitted on 2010 through the previous year and then scored that year, so the result concatenates 12 separately fitted models and no test year appears in its own training data. The window expands each year, so the last year's model saw the most data and the first year's the least. These are the incumbent Offset's margins, reused exactly rather than refitted, so the window is that run's. Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from

these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on is_honest_scale in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+0.069%

Rebate-aware Kelly,
annual reset

BETS

1,682

settled bets

REBATE RECEIVED

HK\$14,000

the most any rule
received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,515	+0.064%	1,671 bets
Kelly, no rebate, continuous	1,515	-0.390%	1,671 bets
Kelly, settled rebate, annual reset	1,516	+0.064%	1,672 bets
Kelly, settled rebate, continuous	1,516	-0.400%	1,672 bets
Rebate-aware Kelly, annual reset	1,516	+0.069%	1,682 bets
Rebate-aware Kelly, continuous	1,516	-0.343%	1,682 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	253	253	254

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2017	284	284	286
2018	231	231	233
2019	160	160	161
2020	193	193	196
2021	149	150	150
2022	141	141	142
2023	79	79	79
2024	81	81	81
2025	80	80	80
2026	20	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 253 bets in 2016 against 80 in 2025. Kelly, settled rebate: 253 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 254 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$14,000; Rebate-aware Kelly, continuous HK\$14,000. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+3.3%

Rebate-aware Kelly,
continuous

BETS

1,835

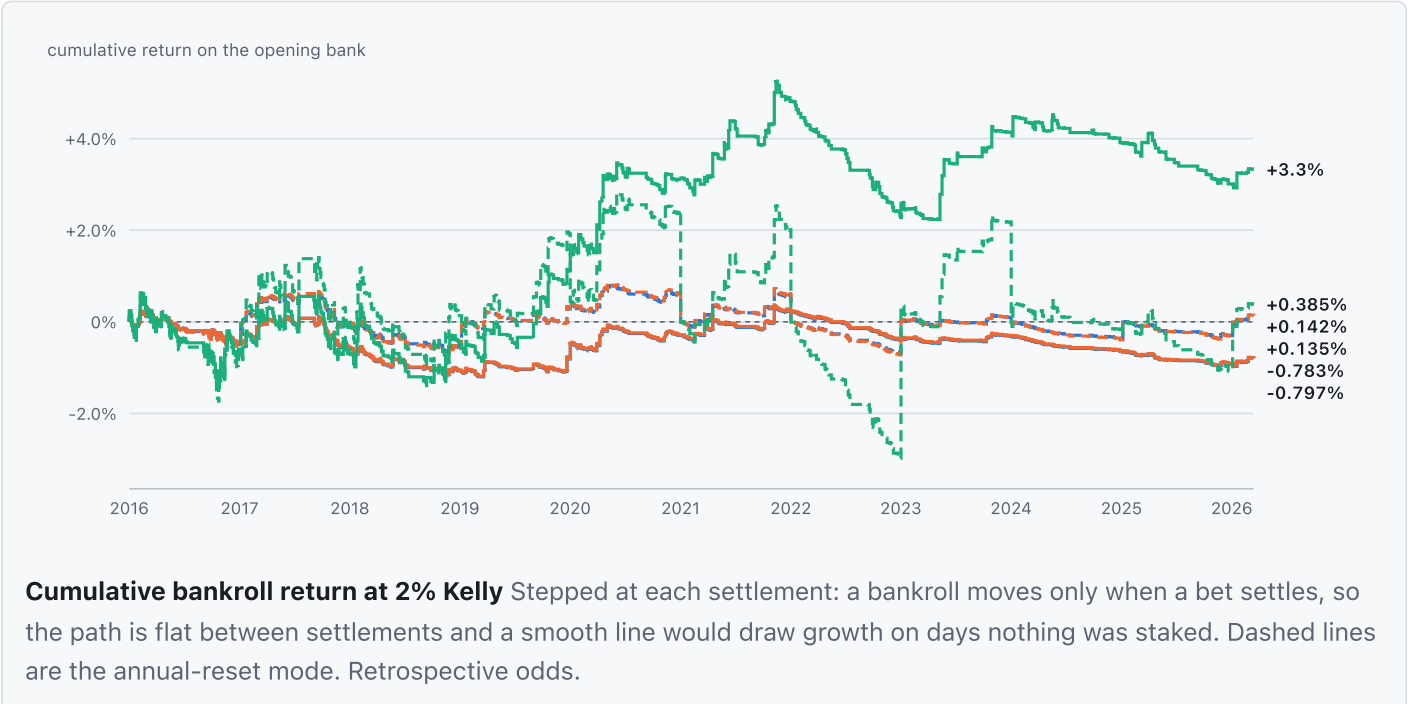
settled bets

REBATE RECEIVED

HK\$315,826

the most any rule
received

■ Kelly, no rebate
■ Kelly, settled rebate
■ Rebate-aware Kelly



▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,537	+0.135%	1,696 bets
Kelly, no rebate, continuous	1,537	-0.797%	1,696 bets
Kelly, settled rebate, annual reset	1,537	+0.142%	1,696 bets
Kelly, settled rebate, continuous	1,537	-0.783%	1,696 bets
Rebate-aware Kelly, annual reset	1,619	+0.385%	1,835 bets
Rebate-aware Kelly, continuous	1,619	+3.3%	1,835 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	254	254	286
2017	287	287	311
2018	233	233	248
2019	162	162	172
2020	198	198	213
2021	153	153	162
2022	144	144	157
2023	82	82	88
2024	83	83	93
2025	80	80	85
2026	20	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 254 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 286 bets in 2016 against 85 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$315,826; Rebate-aware Kelly, continuous HK\$315,826; Kelly, settled rebate, annual reset HK\$1,007; Kelly, settled rebate, continuous HK\$1,007. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+11.8%

Rebate-aware Kelly, continuous

BETS

3,070

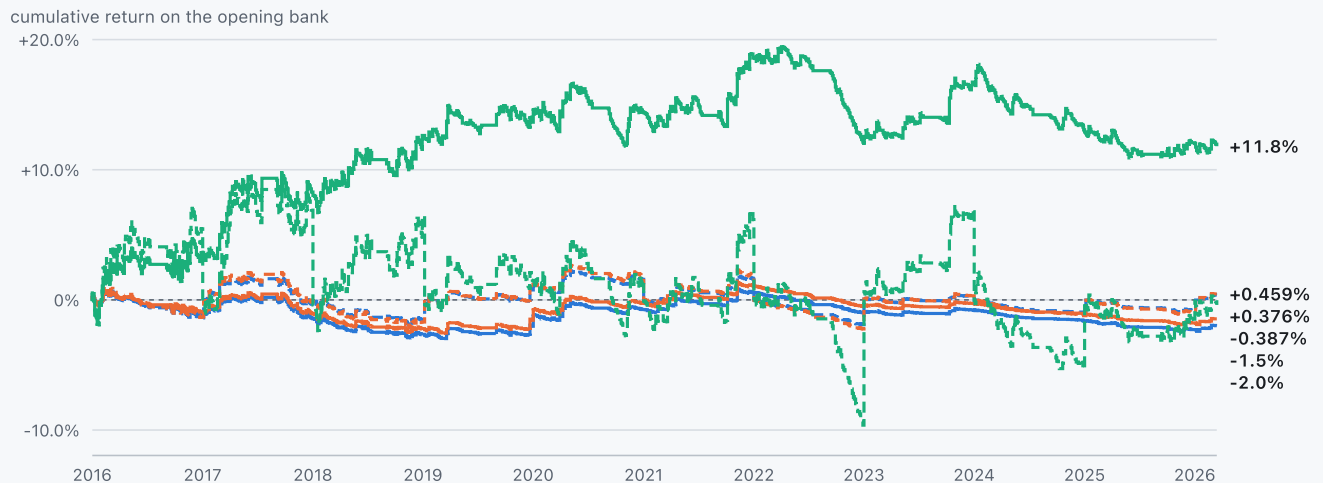
settled bets

REBATE RECEIVED

HK\$2,478,032

the most any rule received

■ Kelly, no rebate
 ■ Kelly, settled rebate
 ■ Rebate-aware Kelly



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,544	+0.376%	1,703 bets
Kelly, no rebate, continuous	1,544	-2.0%	1,703 bets
Kelly, settled rebate, annual reset	1,545	+0.459%	1,704 bets
Kelly, settled rebate, continuous	1,545	-1.5%	1,704 bets
Rebate-aware Kelly, annual reset	2,612	-0.387%	3,070 bets
Rebate-aware Kelly, continuous	2,612	+11.8%	3,070 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	255	255	411
2017	289	290	460
2018	233	233	373
2019	163	163	266
2020	199	199	335
2021	154	154	295
2022	145	145	274
2023	82	82	186
2024	83	83	218
2025	80	80	201
2026	20	20	51

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 255 bets in 2016 against 80 in 2025. Kelly, settled rebate: 255 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 411 bets in 2016 against 201 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$2,478,032; Rebate-aware Kelly, continuous HK\$2,478,032; Kelly, settled rebate, annual reset HK\$75,876; Kelly, settled rebate, continuous HK\$75,876. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is

why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+30.7%

Rebate-aware Kelly,
continuous

BETS

4,596

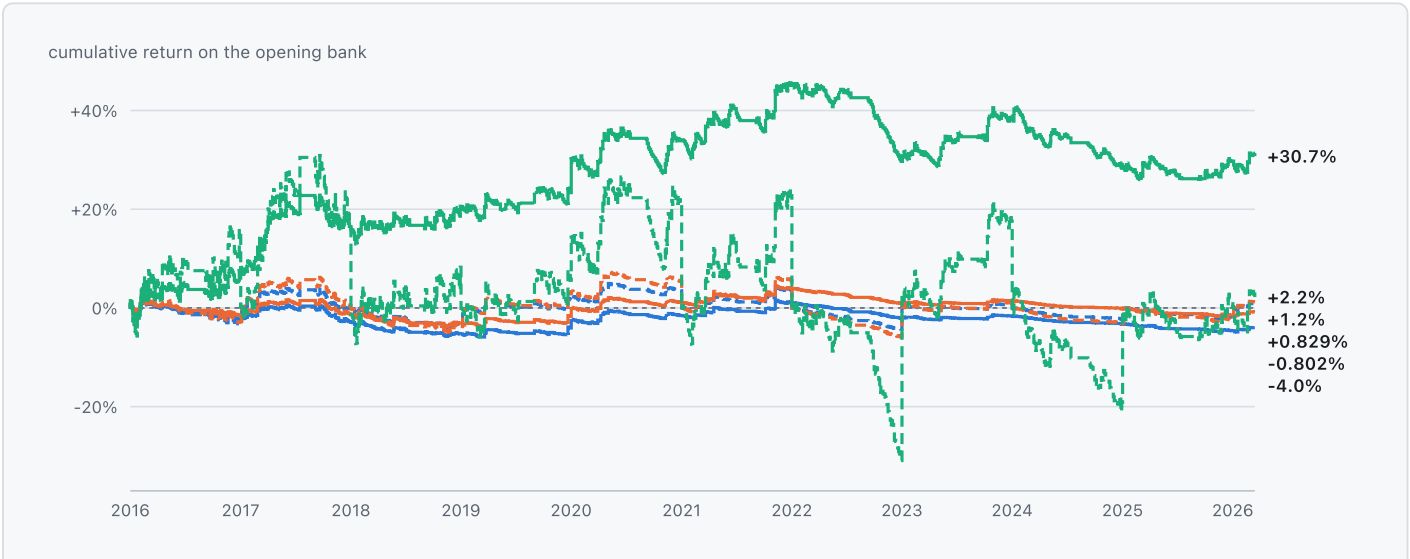
settled bets

REBATE RECEIVED

HK\$8,492,883

the most any rule
received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,545	+0.829%	1,705 bets
Kelly, no rebate, continuous	1,545	-4.0%	1,705 bets
Kelly, settled rebate, annual reset	1,547	+1.2%	1,708 bets
Kelly, settled rebate, continuous	1,547	-0.802%	1,708 bets
Rebate-aware Kelly, annual reset	4,026	+2.2%	4,596 bets
Rebate-aware Kelly, continuous	4,026	+30.7%	4,596 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	255	255	529
2017	290	290	583
2018	234	234	498
2019	163	163	407
2020	199	199	492
2021	154	154	455

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2022	145	146	429
2023	82	82	351
2024	83	83	372
2025	80	80	379
2026	20	22	101

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 255 bets in 2016 against 80 in 2025. Kelly, settled rebate: 255 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 529 bets in 2016 against 379 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$8,492,883; Rebate-aware Kelly, continuous HK\$8,492,883; Kelly, settled rebate, annual reset HK\$638,228; Kelly, settled rebate, continuous HK\$638,228. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,006,394	+0.064%	1,671	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$10,487,212	HK\$10,446,268	-0.390%	1,671	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,006,433	+0.064%	1,672	HK\$0
1% Kelly	Kelly, settled rebate, continuous	HK\$10,642,674	HK\$10,600,136	-0.400%	1,672	HK\$0
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,006,910	+0.069%	1,682	HK\$14,000
1% Kelly	Rebate-aware Kelly, continuous	HK\$10,859,819	HK\$10,822,587	-0.343%	1,682	HK\$14,000
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,013,548	+0.135%	1,696	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$10,956,325	HK\$10,869,024	-0.797%	1,696	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,014,152	+0.142%	1,696	HK\$1,007
2% Kelly	Kelly, settled rebate, continuous	HK\$11,572,600	HK\$11,481,996	-0.783%	1,696	HK\$1,007
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,038,511	+0.385%	1,835	HK\$315,826
2% Kelly	Rebate-aware Kelly, continuous	HK\$11,970,655	HK\$12,369,388	+3.3%	1,835	HK\$315,826
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,037,602	+0.376%	1,703	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$12,254,926	HK\$12,012,729	-2.0%	1,703	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,045,898	+0.459%	1,704	HK\$75,876
5% Kelly	Kelly, settled rebate, continuous	HK\$14,428,220	HK\$14,218,250	-1.5%	1,704	HK\$75,876
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$9,961,312	-0.387%	3,070	HK\$2,478,032
5% Kelly	Rebate-aware Kelly, continuous	HK\$14,916,934	HK\$16,678,772	+11.8%	3,070	HK\$2,478,032
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,082,865	+0.829%	1,705	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$13,911,338	HK\$13,353,997	-4.0%	1,705	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,124,254	+1.2%	1,708	HK\$638,228
10% Kelly	Kelly, settled rebate, continuous	HK\$19,446,802	HK\$19,290,764	-0.802%	1,708	HK\$638,228
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,218,614	+2.2%	4,596	HK\$8,492,883
10% Kelly	Rebate-aware Kelly, continuous	HK\$20,257,323	HK\$26,485,579	+30.7%	4,596	HK\$8,492,883

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA
-0.000844

INTERVAL
-0.0012 to
-0.0004
80% level

EXCLUDES ZERO
FAVOURABLY
yes

RACE DAYS
891
39,665 races

REPLICATES
2,000
seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix

THE UTILITY SCALE

FITTED MINIMUM
1.9932

FITTED MAXIMUM
2.2280

THEORY
0.7797
the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.993 to 2.228 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit_offset
Comparator field	offset
Integration	exact_gauss_hermite_quadrature
Shared factors	0
Opening bankroll	10,000,000

Item	Value
Bankroll mode	continuous
Rebate-aware sizing	no

THE COMPARISON THIS ARCHITECTURE WAS DESIGNED TO MAKE

ISOLATION
COMPARATOR

the incumbent
Offset model on
the same
margins

DELTA

-0.000943

negative is better

CALIBRATED

-0.000493

both sides given a
temperature

YEARS BETTER

11/11

WORST YEAR

-0.000049

The incumbent Offset, because the candidate reuses its margins exactly and changes only the link. The market anchor means a zero correction returns the market probability, so this comparison isolates the link. This report's headline is the market delta, because that is the one quantity all four published models can be compared on. This delta answers a narrower question and answers it better: it is the only comparison that holds everything except the link fixed. Read as a headline it would mislead -- a candidate can beat what it was designed against and still lose to the market -- and read as an isolation result it is exactly what the architecture was built to measure.

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	pass
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

6 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.000947
17	-0.000919
42	-0.000889
123	-0.000943
256	-0.001014

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

UTILITY SCALE

setting	value	note
2015	0.779697	fitted on strictly earlier years; fallback to the theoretical value
2016	1.993223	fitted on strictly earlier years
2017	2.034700	fitted on strictly earlier years
2018	2.210128	fitted on strictly earlier years
2019	2.226151	fitted on strictly earlier years
2020	2.123985	fitted on strictly earlier years
2021	2.160718	fitted on strictly earlier years
2022	2.145303	fitted on strictly earlier years
2023	2.227982	fitted on strictly earlier years
2024	2.216202	fitted on strictly earlier years
2025	2.093658	fitted on strictly earlier years
2026	2.051702	fitted on strictly earlier years

12 row(s) from the sensitivity artifact, in 1 study/studies. Where a study varies an integration setting, a result that moves with it is a property of the integration rather than of the model.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Probit Offset (market-anchored) against the market on the same races. It does not measure this model against the other three except through that shared yardstick.

- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.